

Pairs Trading: Performance of a Relative-Value Arbitrage Rule

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We test a Wall Street investment strategy, “pairs trading,” with daily data over 1962–2002. Stocks are matched into pairs with minimum distance between normalized historical prices. A simple trading rule yields average annualized excess returns of up to 11% for self-financing portfolios of pairs. The profits typically exceed conservative transaction-cost estimates. Bootstrap results suggest that the “pairs” effect differs from previously documented reversal profits. Robustness of the excess returns indicates that pairs trading profits from temporary mispricing of close substitutes. We link the profitability to the presence of a common factor in the returns, different from conventional risk measures.

Wall Street has long been interested in quantitative methods of speculation. One popular short-term speculation strategy is known as “pairs trading.” The strategy has at least a 20-year history on Wall Street and is among the proprietary “statistical arbitrage” tools currently used by hedge funds as well as investment banks. The concept of pairs trading is disarmingly simple. Find two stocks whose prices have moved together historically. When the spread between them widens, short the winner and buy the loser. If history repeats itself, prices will converge and the arbitrageur will profit. It is hard to believe that such a simple strategy, based solely on past price dynamics and simple contrarian principles, could possibly make money. If the U.S. equity market were efficient at all times, risk-adjusted returns from pairs trading should not be positive.

In this article, we examine the risk and return characteristics of pairs trading with daily data over the period 1962 through December 2002.

We are grateful to Peter Bossaerts, Michael Cooper, Jon Ingersoll, Ravi Jagannathan, Maureen O'Hara, Carl Schecter, and two anonymous referees for many helpful discussions and suggestions on this topic. We thank the International Center for Finance at the Yale School of Management for research support, and the participants in the EFA'99 Meetings, the AFA'2000 Meetings, the Berkeley Program in Finance, and the Finance and Economics workshops at Vanderbilt and Wesleyan for their comments. Address correspondence to Evan Gatev, Boston College, Carroll School of Management, Fulton Hall, 140 Commonwealth Ave, Chestnut Hill, MA 02467, or email: gatev@bc.edu.

Using a simple algorithm for choosing pairs, we test the profitability of several straightforward, self-financing trading rules. We find average annualized excess returns of about 11% for top pairs portfolios. Although pairs strategies exploit temporary components of stock prices, we show that our profits are not caused by simple mean reversion as documented in the previous literature. We examine the robustness of our results to a wide variety of risk factors—including not only the widely used factors in the empirical literature but also potential low-frequency institutional factors such as bankruptcy risk. In addition, we explore the robustness of our results to microstructure factors such as the bid-ask bounce, short-selling costs, and transaction costs. Although some factors such as short-selling and transaction costs affect the magnitude of the excess returns, pairs trading remains profitable for reasonable assumptions over the sample period of study, as well as over a true out-of-sample test of four years. We interpret the results of our analysis as evidence in favor of profitable arbitrage in expectations that may accrue to market participants who possess relatively low transaction costs and the ability to short securities. We also find evidence that points to a systematic factor that influences the profitability of pairs trading over time. This unidentified latent risk factor has been relatively dormant recently. The importance of this risk factor is correlated with the returns to pairs trading, which is consistent with the view that the profits are a compensation to arbitrageurs for enforcing the “Law of One Price.”

We argue that our results reveal something about the mechanism and performance of relative-price arbitrage activities in practice. This is potentially useful to researchers because, despite considerable theory about market efficiency, economists have little empirical information about how efficiency is maintained in practice. In addition, despite the fact that hedge funds have attracted an increasing amount of investment capital over the past decade, the study of hedge fund strategies is in its infancy in the financial economics literature. This article examines the risk and return characteristics of one widely practiced active trading strategy.

One natural question to ask is whether our results imply a violation of equilibrium asset pricing. Although the documented profitability of the pairs trading rule is a robust result, it is not inconsistent with all pricing models. Indeed the reversion in relative values we find is consistent with a pricing model in prices developed and tested by Bossaerts (1988). Thus, our article at the very least suggests that this class of models merits further empirical investigation.

The remainder of the article is organized as follows. Section 1 provides some background on pairs trading strategy. The next section describes our methodology of constructing pairs and calculating returns. The empirical results are described in Section 3, and Section 4 provides conclusions and directions for future research.

1. Background of Pairs Trading

1.1 History

In the mid-1980s, the Wall Street quant Nunzio Tartaglia assembled a team of physicists, mathematicians, and computer scientists to uncover arbitrage opportunities in the equities markets. Tartaglia's group of former academics used sophisticated statistical methods to develop high-tech trading programs, executable through automated trading systems, which took the intuition and trader's "skill" out of arbitrage and replaced it with disciplined, consistent filter rules. Among other things, Tartaglia's programs identified pairs of securities whose prices tended to move together. They traded these pairs with great success in 1987—a year when the group reportedly made a \$50 million profit for the firm. Although the Morgan Stanley group disbanded in 1989 after a couple of bad years of performance, pairs trading has since become an increasingly popular "market-neutral" investment strategy used by individual and institutional traders as well as hedge funds. The increased popularity of quantitative-based statistical arbitrage strategies has also apparently affected profits. In a *New York Times* interview, David Shaw, head of one of the most successful modern quant shops and himself an early Tartaglia's protégé, suggests that recent pickings for quant-shops have become slim—he attributes the success of his firm, D. E. Shaw, to early entry into the business. Tartaglia's own explanation for pairs trading is psychological. He claims, "... Human beings don't like to trade against human nature, which wants to buy stocks after they go up not down" [Hansell (1989)]. Could pairs traders be the disciplined investors taking advantage of the undisciplined over-reaction displayed by individual investors? This is at least one possible—albeit psychological—explanation for our results, which is consistent with Jegadeesh and Titman's (1995) finding that contrarian profits are in part due to over-reaction to company-specific information shocks rather than price reactions to common factors.

1.2 Data snooping and market response

In our study we have not searched over the full strategy space to identify successful trading rules, but rather we have interpreted practitioner description of pairs trading as straightforwardly as possible. Our rules follow the general outline of first "find stocks that move together," and second "take a long-short position when they diverge and unwind on convergence." A test requires that both of these steps must be parameterized in some way. How do you identify "stocks that move together?" Need they be in the same industry? Should they only be liquid stocks? How far do they have to diverge before a position is put on? When is a position unwound? We have made some straightforward choices about each of these questions. We put positions on at a two-standard deviation

spread, which might not always cover transaction costs even when stock prices converge. Although it is tempting to try potentially more profitable schemes, the danger in data-snooping refinements outweigh the potential insights gained about the higher profits that could result from learning through testing.¹

As with all filter rules using historical asset pricing data, data snooping is a potential concern. One approach to the data snooping issue is to test the results out of sample. We completed and circulated the first draft of the working paper in 1999, using data through the end of 1998. The time lag between the first analysis and the present study gives us an ideal holdout sample. Using the original model, but the post-1988 data, we found that over the 1999–2002 period, the excess return of the fully invested portfolio of the top 20 pairs averaged 10.4% per annum, with an annual standard deviation of 3.8% and a large and significant Newey-West-adjusted *t*-statistic of 4.82—consistent with the long-term, in-sample results of our original analysis. We were careful not to adjust our strategy from the first draft to the current draft of the article, to avoid data-snooping criticisms. Not only does this additional four-year sample suggest that the results were not simply an artifact of the earlier sample period, over which pairs trading was known to be popular, but it also suggests that the public dissemination of the results has apparently not affected the general risk and return characteristics of the strategy, despite curiosity from the professional sector.

1.3 Relative pricing

Asset pricing can be viewed in absolute and relative terms. Absolute pricing values securities from fundamentals such as discounted future cash flow. This is a notoriously difficult process with a wide margin for error. Articles by Bakshi and Chen (1997) and Lee et al. (1997), for example, are heroic attempts to build quantitative value-investing models. Relative pricing is only slightly easier. Relative pricing means that two securities that are close substitutes for each other should sell for the same price—it does not say what that price will be. Thus, relative pricing allows for bubbles in the economy, but not necessarily arbitrage or profitable speculation. The Law of One Price [LOP] and a “near-LOP” are applicable to relative pricing—even if that price is wrong.

Ingersoll (1987) defines the LOP as the “proposition ... that two investments with the same payoff in every state of nature must have the same current value.” In other words, two securities with the same prices in all states of the world should sell for the same amount. Chen and Knez (1995) extend this to argue that “*closely integrated* markets should assign

¹ Froot and Dabora (1999) consider “twin” stocks that trade in different international markets to examine the issues of market integration.

to similar payoffs prices that are *close*.” They argue that two securities with similar, but not necessarily, matching payoffs across states should have similar prices. This is of course a weaker condition and subject to bounds on prices for unusual states; however, it allows the examination of “near-efficient” economies, or in Chen and Knez’ case, near integrated markets. Notice that this theory corresponds to the desire to find two stocks whose prices move together as long as we can define states of nature as the time series of observed historical trading days.

We use an algorithm to choose pairs based on the criterion that they have had the same or nearly the same state prices historically. We then trade pairs whose prices closely match in historical state-space, because the LOP suggests that in an efficient market their prices should be nearly identical. In this framework, the current study can be viewed as a test of the LOP and near-LOP in the U.S. equity markets, under certain stationarity conditions. We are effectively testing the integration of very local markets—the markets for specific individual securities. This is similar in spirit to Bossaerts’ (1988) test of co-integration of security prices at the portfolio level. We further conjecture that the marginal profits to be had from risk arbitrage of these temporary deviations is crucial to the maintenance of first-order efficiency. We could not have the first effect without the second.

1.4 Co-integrated prices

The pairs trading strategy may be justified within an equilibrium asset-pricing framework with nonstationary common factors like Bossaerts and Green (1989) and Jagannathan and Viswanathan (1988). If the long and short components fluctuate with common nonstationary factors, then the prices of the component portfolios would be co-integrated and the pairs trading strategy would be expected to work. Evidence of exposures to common nonstationary factors would support a nonstationary factor pricing framework.

The space of normalized, cum-dividend prices, that is, cumulative total returns with dividends reinvested, is the basic space for the pairs trading strategies in this article. The main observation about our motivating models of the CAPM-APT variety is that they are known to imply perfect collinearity of prices, which is readily rejected by the data. On the other hand, Bossaerts (1988) finds evidence of price co-integration for the U.S. stock market. We would like to keep the notion of the empirically observed co-movement of prices, without unnecessarily restrictive assumptions, hence we proceed in the spirit of the co-integrated prices literature. More specifically, our matching in price space can be interpreted as follows. Suppose that prices obey a statistical model of the form,

$$p_{it} = \sum \beta_{ik} p_{kt} + \varepsilon_{it}, k < n \quad (1)$$

where ε_{it} denotes a weakly dependent error in the sense of Bossaerts (1988). Assume also that p_{it} is weakly dependent after differencing once. Under these assumptions, the price vector $\mathbf{p}_t$ is co-integrated of order 1 with co-integrating rank $r = n - k$, in the sense of Engle and Granger (1987) and Bossaerts (1988). Thus, there exist r linearly independent vectors $\{\alpha_q\}_q = 1, \dots, r$ such that $z_q = \alpha_q' \mathbf{p}_t$ are weakly dependent. In other words, r linear combinations of prices will not be driven by the k common nonstationary components p_t . Note that this interpretation does not imply that the market is inefficient, rather it says that certain assets are weakly redundant, so that any deviation of their price from a linear combination of the prices of other assets is expected to be temporary and reverting.

To interpret the pairs as co-integrated prices, we need to assume that for $n \gg k$, there are co-integrating vectors that have only two nonzero coordinates. In that case, the sum or difference of scaled prices will be reverting to zero and a trading rule could be constructed to exploit the expected temporary deviations. Our strategy relies on exactly this conclusion. In principle one could construct trading strategies with trios, quadruples, and so on of stocks, which would presumably capture more co-integrated prices and would yield better profits.

The assumption that a linear combination of two stocks can be weakly dependent may be interpreted as saying that a co-integrating vector can be partitioned in two parts, such that the two corresponding portfolios are priced within a weakly dependent error of another stock. Given the large universe of stocks, this statement is always empirically valid and provides the basis of our formation procedure.² However, it is important to recognize the possibility of spuriously correlated prices, which are not de facto co-integrated.

1.5 Bankruptcy risk

The risk of bankruptcy is one reason why the returns on individual securities cannot be taken as stationary. Sensitivity of the pairs trading to the default premium suggests that the strategy may work because we are pairing two firms, the first of which may have a constant or decreasing probability of bankruptcy (short end), whereas the second may have a temporarily increasing probability of bankruptcy (long end). The “surprise improvements” in the short end are then followed by improvement in the long end if that stock survives. In other words, the source of the profit is the improving ex post (non)realization of idiosyncratic bankruptcy risk in the long (loser) stock. In such case, we would expect to have asymmetry in the profits from the long and the short components, with

² Note that the case $n \gg k$ corresponds to the standard finance paradigm where in the large universe of n stocks, expected returns are driven by a few, namely k , common factors. This paradigm is supported by existing empirical work, for example, see Connor and Korajczyk (1993) for references, which generally finds less than 10 common nonstationary components.

most of the profits coming from the long end.³ We test long and short positions separately to see if this is driving our results.

2. Methodology

Our implementation of pairs trading has two stages. We form pairs over a 12-month period (formation period) and trade them in the next 6-month period (trading period). Both 12 months and 6 months are chosen arbitrarily and have remained our horizons since the beginning of the study.

2.1 Pairs formation

In each pairs formation period, we screen out all stocks from the CRSP daily files that have one or more days with no trade. This serves to identify relatively liquid stocks as well as to facilitate pairs formation. Next, we construct a cumulative total returns index for each stock over the formation period. We then choose a matching partner for each stock by finding the security that minimizes the sum of squared deviations between the two normalized price series. Pairs are thus formed by exhaustive matching in normalized daily “price” space, where price includes reinvested dividends.

We use this approach because it best approximates the description of how traders themselves choose pairs. Interviews with pairs traders suggest that they try to find two stocks whose prices “move together.” In addition to “unrestricted” pairs, we will also present results by sector, where we restrict both stocks to belong to the same broad industry categories defined by Standard and Poors: Utilities, Transportation, Financial, and Industrials. Each stock is assigned to one of these four groups, based on the stock’s SIC code. The minimum-distance criterion is then used to match stocks within each of the groups.

2.2 Trading period

Once we have paired up all liquid stocks in the formation period, we study the top 5 and 20 pairs with the smallest historical distance measure, in addition to the 20 pairs after the top 100 (pairs 101–120). This last set is valuable because most of the top pairs share certain characteristics, which will be described in detail below. On the day following the last day of the pairs formation period, we begin to trade according to a prespecified rule.

Figure 1 illustrates the pairs trading strategy using two stocks, Kennecott and Uniroyal, in the six-month period starting in August of 1962. The top two lines represent the normalized price paths with dividends reinvested and the bottom line indicates the opening and closing of the strategy on a daily basis. It is clear why these two firms paired with each other. They generally tended to move together over the trading interval.

³ We thank an anonymous referee for this example.

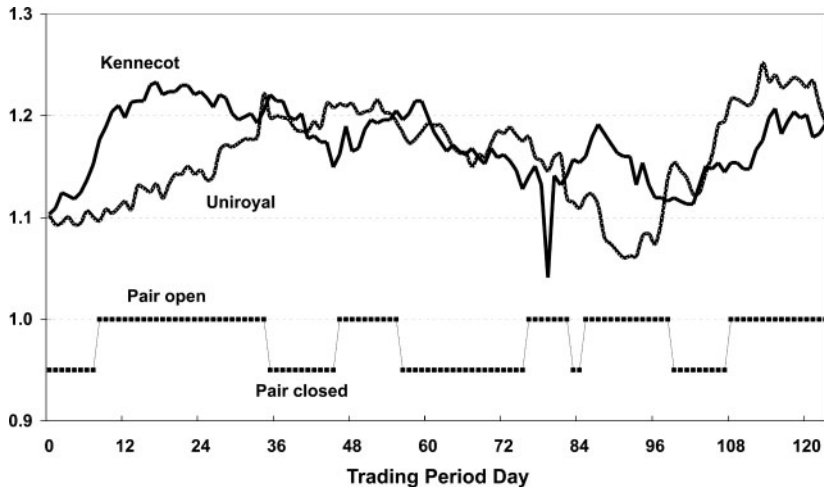


Figure 1
Daily normalized prices: Kennecott and Uniroyal (pair 5)
Trading period August 1963–January 1964.

Notice that the position first opens in the seventh trading day of the period and then remains open until day 36. Over that interval, the spread actually first increased significantly before convergence. The prices remain close during the period and cross frequently. The pair opens five times during the period, however not always in the same direction. Neither stock is the “leader.” In our example, convergence occurs in the final day of the period, although this is not always the case.

We select trading rules based on the proposition that we open a long–short position when the pair prices have diverged by a certain amount and close the position when the prices have reverted. Following practice, we base our rules for opening and closing positions on a standard deviation metric. We open a position in a pair when prices diverge by more than two historical standard deviations, as estimated during the pairs formation period. We unwind the position at the next crossing of the prices. If prices do not cross before the end of the trading interval, gains or losses are calculated at the end of the last trading day of the trading interval. If a stock in a pair is delisted from CRSP, we close the position in that pair, using the delisting return, or the last available price.⁴ We report the

⁴ The profits are robust with respect to this delisting assumption. A potential problem arises if inaccurate and stale prices exaggerate the excess returns and bias the estimated return of a long position in a plummeting stock. To address this potential concern, we have reestimated our results under the extreme assumption that only a *long* stock experiences a –100% return when it is delisted. This zero-price extreme includes, among other things, the possibility of nontrading due to the lack of liquidity. Because selective loss on the long position always harms the pair profit, this extreme assumption biases the results against profitability. However, pairs trading remains profitable under this alternative: for example, the average monthly return on the top 20 pairs portfolio is 1.32% with a standard deviation of 1.9%.

payoffs by going one dollar short in the higher-priced stock and one dollar long in the lower-priced stock.

2.3 Excess return computation

Because pairs may open and close at various points during the six-month trading period, the calculation of the excess return on a portfolio of pairs is a nontrivial issue. Pairs that open and converge during the trading interval will have positive cash flows. Because pairs can reopen after initial convergence, they can have multiple positive cash flows during the trading interval. Pairs that open but do not converge will only have cash flows on the last day of the trading interval when all positions are closed out. Therefore, the payoffs to pairs trading strategies are a set of positive cash flows that are randomly distributed throughout the trading period, and a set of cash flows at the end of the trading interval that can be either positive or negative. For each pair we can have multiple cash flows during the trading interval, or we may have none in the case when prices never diverge by more than two standard deviations during the trading interval. Because the trading gains and losses are computed over long–short positions of one dollar, the payoffs have the interpretation of excess returns. The excess return on a pair during a trading interval is computed as the reinvested payoffs during the trading interval.⁵ In particular, the long and short portfolio positions are marked-to-market daily. The daily returns on the long and short positions are calculated as value-weighted returns in the following way,

$$r_{P,t} = \frac{\sum_{i \in P} w_{i,t} r_{i,t}}{\sum_{i \in P} w_{i,t}} \quad (2)$$

$$w_{i,t} = w_{i,t-1}(1 + r_{i,t-1}) = (1 + r_{i,1}) \cdots (1 + r_{i,t-1}) \quad (3)$$

where r defines returns and w defines weights, and the daily returns are compounded to obtain monthly returns. This has the simple interpretation of a buy-and-hold strategy.

We consider two measures of excess return on a *portfolio* of pairs: the return on committed capital and the fully invested return, that is, the return on actual employed capital. The former scales the portfolio payoffs by the number of pairs that are selected for trading, the latter divides the payoffs by the number of pairs that open during the trading period. The former measure of excess return is clearly more conservative: if a pair does not trade for the whole of the trading period, we still include a dollar

⁵ This is a conservative approach to computing the excess return, because it implicitly assumes that all cash earns zero interest rate when not invested in an open pair. Because any cash flow during the trading interval is positive by construction, it ignores the fact that these cash flows are received early and understates the computed excess returns.

of committed capital as the cumulative return in our calculation of excess return. It takes into account the opportunity cost of hedge funds of having to commit capital to a strategy even if the strategy does not trade. To the extent that hedge funds are flexible in their sources and uses of funds, computing excess return relative to the actual capital employed may give a more realistic measure of the trading profits.

We initiate the pairs strategy by trading the pairs at the beginning of every month in the sample period, with the exception of the first 12 months, which are needed to estimate pairs for the strategy starting in the first month. The result is a time series of overlapping six-month trading period excess returns. We correct for the correlation induced by overlap by averaging monthly returns across trading strategies that start one month apart as in Jegadeesh and Titman (1993). The resulting time series has the interpretation of the payoffs to a proprietary trading desk, which delegates the management of the six portfolios to six different traders whose formation and trading periods are staggered by one month.

3. Empirical Results

3.1 Strategy profits

Table 1 summarizes the excess returns for the pairs portfolios that are unrestricted in the sense that the matching stocks do not necessarily belong to the same broad industry categories. In Section 3.5 we will consider sector-neutral pairs strategies. Panel A summarizes the excess returns of pairs strategies when positions are opened at the end of the day that prices diverge and closed at the end of the day of price convergence. The first row shows that a fully invested portfolio of the five best pairs earned an average excess monthly return of 1.31% (t -statistic = 8.84), and a portfolio of the 20 best pairs 1.44% per month (t = 11.56). Using the more conservative approach to computing excess returns, using committed capital, gives excess returns of 0.78 and 0.81% per month, respectively. Either way, these excess returns are large in an economical and statistical sense and suggest that pairs trading is profitable.

The remainder of Panel A provides information about the excess return distributions of pairs portfolios. There are diversification benefits from combining multiple pairs in a portfolio. As the number of pairs in a portfolio increases, the portfolio standard deviation falls. The diversification benefits are also apparent from the range of realized returns. Interestingly, as the number of pairs in the strategy increases, the minimum realized return increases, whereas the maximum realized excess return remains relatively stable. During the full sample period of 474 months, a portfolio of 20 pairs experienced 71 monthly periods with negative payoffs, compared to 124 months for a portfolio of 5 pairs. The decrease in

Table 1
Excess returns of unrestricted pairs trading strategies

Pairs portfolio	Top 5	Top 20	101–120	All
A. Excess return distribution (no waiting)				
Average excess return (fully invested)	0.01308	0.01436	0.01081	0.01104
Standard error (Newey-West)	0.00148	0.00124	0.00094	0.00099
<i>t</i> -Statistic	8.84	11.56	11.54	11.16
Excess return distribution				
Median	0.01194	0.01235	0.00955	0.00728
Standard deviation	0.02280	0.01688	0.01540	0.01670
Skewness	0.62	1.39	1.34	3.42
Kurtosis	7.81	10.54	10.30	25.25
Minimum	−0.10573	−0.06629	−0.03857	−0.02721
Maximum	0.14716	0.13295	0.12684	0.17178
Observations with excess return < 0	26%	15%	21%	17%
Average excess return on committed capital	0.00784	0.00805	0.00679	0.00614
B. Excess return distribution (one day waiting)				
Average monthly return (fully invested)	0.00745	0.00895	0.00795	0.00715
Standard error (Newey-West)	0.00119	0.00096	0.00085	0.00090
<i>t</i> -Statistic	6.26	9.29	9.40	7.92
Excess return distribution				
Median	0.00699	0.00690	0.00694	0.00411
Standard deviation	0.02101	0.01527	0.01438	0.01577
Skewness	0.34	1.45	0.98	3.32
Kurtosis	10.64	16.13	7.78	25.66
Minimum	−0.12628	−0.08218	−0.04266	−0.02951
Maximum	0.14350	0.13490	0.10464	0.16325
Observations with excess return < 0	35%	23%	28%	32%
Average excess return on committed capital	0.00463	0.00520	0.00503	0.00396

Summary statistics of the monthly excess returns on portfolios of pairs between July 1963 and December 2002 (474 observations). We trade according to the rule that opens a position in a pair at the end of the day that prices of the stocks in the pair diverge by two historical standard deviations (Panel A). The results in Panel B correspond to a strategy that delays the opening of the pairs position by one day. All pairs are ranked according to least distance in historical price space. The “top *n*” portfolios include the *n* pairs with least distance measures, and the portfolio “101–120” studies the 20 pairs after the top 100. The average number of pairs in the all-pair portfolio is 2057. The *t*-statistics are computed using Newey-West standard errors with six-lag correction. Absolute kurtosis is reported.

the standard deviation and the increase of the lower end of the return distribution are also reflected in an increased skewness coefficient.

Because pairs trading is in essence a contrarian investment strategy, the returns may be biased upward because of the bid-ask bounce [Jegadeesh (1990), Jegadeesh and Titman (1995), Conrad and Kaul (1989)]. In particular, our strategy sells stocks that have done well relative to their match and buys those that have done poorly. Part of any observed price divergence is potentially due to price movements between bid and ask quotes: conditional on divergence, the winner’s price is more likely to be an ask quote and the loser’s price a bid quote. In Panel A we have used these same prices for the start of trading and our returns may be biased upward because of the fact that we are implicitly buying at bid quotes (losers) and selling at ask quotes (winners). The opposite is true at the second crossing

(convergence): part of the drop in the winner's price can reflect a bid quote, and part of the rise of the loser's price—an ask quote.

To address this issue, Panel B of Table 1 provides the excess returns when we initiate positions in each pair on the day following the divergence and liquidate on the day following the crossing. The average excess returns on the fully invested portfolios and on committed capital drop by about 30–55 and 20–35 basis points (bp), respectively. Although the excess returns remain significantly positive, the drop in excess returns suggests that a nontrivial portion of the profits in Panel A may be due to bid–ask bounce. It is difficult to quantify which portion of the profit reduction is due to bid–ask bounce and which portion stems from true mean reversion in prices because of rapid market adjustment. Nonetheless, this difference raises questions about the economic significance of our results when we include transaction costs. We will return to a detailed discussion of this issue in Section 3.3. Unless stated otherwise, the remainder of the article will report results for pairs strategies that open (close) on the day following divergence (convergence).

3.2 Trading statistics and portfolio composition

Table 2 summarizes the trading statistics and composition of the pairs portfolios. What are the characteristics of the stocks that are matched into pairs? How often does a typical pair trade? Because pairs trading is an active investment strategy, it is important to evaluate the profitability relative to the trading intensity of the portfolios. As mentioned before, we use a two standard deviation trigger to open a pairs position. The second line of panel A in Table 2 reports the average price deviation of the two standard deviation trigger. For the top five pairs, the position typically opens when prices have diverged by 4.76% or more. This is a relatively narrow gap in prices.⁶ The trigger spread increases with the number of pairs in the portfolio, because the standard deviation of the prices increases as the proximity of the securities in price space decreases. The next lines of Panel A also shows that on average almost all pairs open during the six-month trading period, and on average more than once. Of the top 5 pairs, on average 4.81 open during the trading period, and the average number of round trips per pair is 2.02. The average duration of an open position is 3.75 months. This indicates that pairs trading—implemented according to the particular rules we chose—is a medium-term investment strategy.

Panel B of Table 2 describes the composition of the pairs in terms of market capitalization and industry membership. In terms of size, the average stock in the top 5 and top 20 pairs belongs to the second and

⁶ The optimal trigger point in terms of profitability may actually be much higher than two standard deviations, although we have not experimented to find out.

Table 2
Trading statistics and composition of pairs portfolios

Pairs portfolio	Top 5	Top 20	101–120	All
A. Trading statistics				
Average price deviation trigger for opening pairs	0.04758	0.05284	0.07560	0.16888
Average number of pairs traded per six-month period	4.81	19.30	19.41	1944.22
Average number of round-trip trades per pair	2.02	1.96	1.78	1.62
Standard deviation of number of round trips per pair	0.62	0.40	0.27	0.16
Average time pairs are open in months	3.75	3.76	3.98	3.97
Standard deviation of time open, per pair, in months	0.80	0.45	0.38	0.17
B. Pairs portfolio composition				
Average size decile of stocks	2.54	2.71	3.41	4.57
Average weight of stocks in top three size deciles	0.78	0.74	0.58	0.40
Average weight of stocks in top five size deciles	0.91	0.91	0.79	0.62
Average weight of pairs from different deciles	0.66	0.69	0.75	0.82
Average decile difference for mixed pairs	0.97	0.97	0.97	0.98
Average sector weights				
Utilities	0.72	0.71	0.32	0.08
Transportation	0.02	0.02	0.02	0.03
Financials	0.11	0.13	0.26	0.16
Industrials	0.15	0.14	0.40	0.73
Mixed sector pairs	0.20	0.22	0.44	0.33

Trading statistics and composition of portfolios of pairs portfolios between July 1963 and December 2002 (474 months). Pairs are formed over a 12-month period according to a minimum-distance criterion and then traded over the subsequent 6-month period. We trade according to the rule that opens a position in a pair on the day following the day on which the prices of the stocks in the pair diverge by two historical standard deviations. The “top n ” portfolios include the n pairs with least distance measures, and the portfolio “101–120” includes the 20 pairs after the top 100. Panel A summarizes the trading characteristics of a pairs strategy. Pairs are opened when prices diverge by two standard deviations. Average deviation to trigger opening of pair is the cross-sectional average of two standard deviations of the pair prices difference. Panel B contains information about the size and industry membership of the stocks in the various pairs portfolios.

third deciles from the top; 74% of the stocks in the top 20 pairs belong to the top three size deciles using CRSP breakpoints, and 91% come from the top five size deciles. About two-thirds of the pairs combine stocks from different size deciles (i.e., “size mixed pairs”), and the stocks in mixed pairs differ on average by a single decile.

The remainder of Panel B gives a breakdown of the pairs by industry composition. On average, 71% of the stocks in the top 20 pairs are utility stocks, despite the fact that Utilities represent a fairly small proportion of the stocks in the whole sample. This is not surprising perhaps because utility stocks tend to have lower volatility and tend to be correlated with interest rate innovations. The strategy does not always match stocks within sectors. The percentage of mixed sector pairs ranges from 20% for the top five pairs to 44% for pairs 101–120. Given the predominance of utilities among the top pairs, it is fair to ask whether the profitability of pairs trading profitability is limited to the utility sector, or whether pairs strategies are also profitable in other sectors of the market. We address this question in Section 3.5.

3.3 Transaction costs

Table 1 summarizes that the average monthly excess return of unrestricted pairs strategies falls from 1.44%, for the top 20 portfolio, to 0.90% per month if we postpone the trades to the day following the crossing. This drop in the excess returns implies an estimate of the average bid-ask spread and hence the transaction costs of trading in the sample. Although actual transaction costs may be different, it is informative to know whether the trading profits are large enough to survive this estimate of transaction costs.

Suppose the extreme case where the prices of the winner at the first crossing (divergence) are ask prices and the loser are bid prices. If the next day prices are equally likely to be at bid or ask, then delaying trades by one day will reduce the excess returns on average by half the sum of the spreads of the winner and the loser. If at the second crossing (convergence) of the pairs the winner is trading at the bid and the loser at the ask, waiting one day will reduce the excess returns on average again by one-half of the sum of the bid-ask spreads of both stocks. In this extreme case, waiting a day before trading reduces the return on each pair by the round-trip transaction costs in that pair. Because we trade each pair on average two times during the six-month trading interval, the drop in the excess returns of 324 bp per six months by waiting one day reflects the cost of two round trips, which implies a transaction cost of 162 bp per pair per round trip. This may be interpreted as an estimated effective spread of 81 bp. The effective spread for the all-pair portfolio is 70 bp. This indirect estimate is higher than the transaction costs reported by Peterson and Fialkowski (1994), who find that the average effective spread for stocks in the CRSP database in 1991 was 37 bp, and is consistent with the trading costs estimated by Keim and Madhavan (1997). Because 91% of the stocks in the top 20 pairs belong to the top five deciles of CRSP stocks, it is possible that the effective spread is even lower than 37 bp.

Do our trading strategies survive these transaction costs? The profits on our trading strategies in Table 1 range from 437 to 549 bp over a six-month period. If the prices used to compute these excess returns are equally likely to be at bid or ask, which seems a reasonable assumption, we have to correct these excess returns to reflect that in practice we buy at the ask and sell at the bid prices. In other words, we have to subtract the round-trip trading costs to get an estimate of the profits after transaction costs. Our conservative estimate of transaction costs of 162 bp times two round trips per pair results in an estimate of 324 bp transaction cost per pair per six-month period. This gives average net profits ranging from 113 to 225 bp over each six-month period. Comparing these profits to the reported standard errors, we conclude that they are both economically and statistically significant.

Further analysis is required to get more precise estimates of influence of transaction costs of pairs trading strategies. An important question in this context is whether the trading rule that we have used to open and close pairs can be expected to generate economically significant profits *even if pairs trading works perfectly*. Because we use a measure of historical standard deviation to trigger the opening of pairs, and because this estimated standard deviation is the smallest among all pairs, it is likely to underestimate the true standard deviation of a pair. As a consequence, we may simply be opening pairs “too soon” and at a point that we cannot expect it to compensate for transaction costs even if the pair subsequently converges. Results that are not reported here suggest that this is indeed the case for some of our pairs.

There is a second reason why our trading strategies require “too much” trading. We open pairs at any point during the trading period when the normalized prices diverge by two standard deviations. This is not a sensible rule toward the end of a trading interval. For example, suppose that a divergence occurs at the next to last day of the trading interval. The convergence has to be substantial to overcome the transaction cost that will be incurred when we close out the position on the next day (the last day of the trading interval). Unreported results suggest that this is also an important source of excess trading.

3.4 Pairs trading by industry group

The pairs formation process thus far has been entirely mechanical. A computer stock has the opportunity to match with a steel firm and a utility with a bank stock. This does not mean that these matches are likely. As summarized in Table 2, the fraction of mixed pairs is typically well below 50%. Common factor exposures of stocks in the same industry will make it more likely to find a match within the same sector. Also, firms that are in industries where cross-sectional differences in factor exposures are small or return variances are low are more likely to end up among the top ranking of pairs. For this reason it is perhaps not surprising that many of the top pairs match two utilities. Are the profits to pairs trading consistent across sectors? We examine the returns on pairs trading where stocks are matched only within the four large sector groupings used by Standard and Poor's: Utilities, Transportation, Financials, and Industrials. The results are summarized in Table 3. As in Table 1, the pairs are traded with a one-day delay before opening and closing a position to minimize the effect of the bid-ask bounce on trading. The monthly excess returns for the top 20 pairs are the largest in the Utilities sector, with 1.08% (Newey-West $t = 10.26$). The profits for the other industry groups are somewhat lower, but all statistically significant, with the average Transportation, Financials, and Industrials top 20 pairs earning 0.58% (Newey-West $t = 4.26$), 0.78% (Newey-West

Table 3
Industry sector pairs trading

Portfolio	Top 5	Top 20	20 after 100	All
A. Utilities				
Mean excess return	0.00905	0.01084	0.009256	0.01036
<i>t</i> -Statistic (Newey-West)	7.37	10.26	6.11	10.51
Median	0.00829	0.00938	0.00665	0.00969
Standard deviation	0.02154	0.01645	0.02640	0.01472
Skewness	0.44	0.76	0.66	1.39
Kurtosis	13.67	12.38	5.52	12.74
Minimum	-0.12347	-0.08750	-0.07868	-0.03519
Maximum	0.16563	0.12730	0.12133	0.12878
Observations with excess return < 0	28%	19%	35%	18%
B. Transportation				
Mean excess return	0.00497	0.00577		0.00440
<i>t</i> -Statistic (Newey-West)	2.98	4.26		3.07
Median	0.00594	0.00547		0.00339
Standard deviation	0.03892	0.02942		0.02871
Skewness	-0.50	-0.11		0.53
Kurtosis	4.98	5.00		5.89
Minimum	-0.19570	-0.12467		-0.12526
Maximum	0.10961	0.11099		0.13560
Observations with excess return < 0	44%	42%		44%
C. Financials				
Mean excess return	0.00678	0.00775	0.00854	0.00726
<i>t</i> -Statistic (Newey-West)	4.79	7.60	7.12	7.62
Median	0.00557	0.00615	0.00660	0.00511
Standard deviation	0.02598	0.01792	0.02358	0.01808
Skewness	0.81	0.61	1.98	2.64
Kurtosis	6.37	6.48	18.27	21.21
Minimum	-0.081988	-0.07810	-0.100477	-0.04452
Maximum	0.149291	0.10638	0.195797	0.17211
Observations with excess return < 0	40%	33%	32%	34%
D. Industrial				
Mean excess return	0.00490	0.00607	0.00664	0.00715
<i>t</i> -Statistic (Newey-West)	4.11	6.93	6.80	7.52
Median	0.00292	0.00547	0.00550	0.00397
Standard deviation	0.02361	0.01601	0.01825	0.01672
Skewness	0.40	0.43	0.94	2.97
Kurtosis	4.94	4.75	7.65	20.95
Minimum	-0.10016	-0.04786	-0.05412	-0.03431
Maximum	0.10546	0.08022	0.11619	0.16072
Observations with excess return < 0	42%	36%	35%	34%

Summary statistics for the excess monthly return distributions for pairs trading portfolios by sector. We trade according to the "wait one day" rule described in the text. The average number of stocks in the industry groups are as follows: 156 Utilities, 61 Transportation, 371 Financials, and 1729 Industrials. There is no "20 after 100" portfolio for the Transportation industry group. The *t*-statistic of the mean is computed using Newey-West standard errors with six lags.

$t = 7.60$), and 0.61% (Newey-West $t = 6.93$), respectively, over a one-month period.

Table 3 also gives a more detailed picture of the return distributions and trading characteristics of the pairs trading strategies by sector. It shows that

the excess return distributions of the sector pairs portfolios are generally skewed right and exhibit positive excess kurtosis relative to a normal distribution. The conclusion from these tables is that pairs trading is profitable in every broad sector category, and not limited to a particular sector.

3.5 The risk characteristics of pairs trading strategies

To provide further perspective on the risk of pairs trading, Table 4 compares the risk-premium of pairs trading to the market premium (S&P 500) and reports the risk-adjusted returns to pairs trading using two different models for measuring risk. Table 5 summarizes value-at-risk (VAR) measures for pairs portfolios.

The top part of Table 4 compares the excess return to pairs trading to the excess return on the S&P 500. Between 1963 and 2002, the average excess return to pairs trading has been about twice as large as the excess return of the S&P 500, with only one-half to one-third of the risk as measured by standard deviation. As a result, the Sharpe Ratios of pairs trading are between four and six times larger than the Sharpe Ratio of the market. Goetzmann et al. (2002) show that Sharpe Ratios can be misleading when return distributions have negative skewness. This is unlikely to be a concern for our study, because our Table 1 showed that the returns to pairs portfolios are positively skewed, which—if anything—would bias our Sharpe Ratios downward.

To explore the systematic risk exposure of the pairs portfolios, we regress their monthly excess returns on the three factors of Fama and French (1996), augmented by two additional factors. The motivation for the additional factors is that pairs strategies invest based on the relative strength of individual stocks. It is therefore possible that pairs trading simply exploits patterns in returns that are known to earn significant profits. For example, Jegadeesh (1990) and Lehmann (1990) show that reversal strategies that select stocks based on prior one-month return earn positive abnormal returns. We control for this possibility by constructing a short-term reversal factor measured as the excess return of stocks in the top three deciles of prior-month return minus the return on stocks in the bottom three deciles.⁷ If pairs strategies sell short-term winners and buy short-term losers, we expect the exposures of pairs portfolios to be positive to the reversal factor. The second additional factor controls for exposure to medium-term return continuation [Jegadeesh and Titman (1993)]. To the extent that pairs trading sells medium-term winners and buys medium-term losers, the pairs excess returns will be negatively correlated with momentum. To examine this possibility, we include a

⁷ The construction is similar to Carhart's (1997) momentum factor, but the performance-sorting horizon here is one month.

Table 4
Systematic risk of pairs trading strategies

	Top 5	Top 20	20 after top 100	All	Equity premium
“Wait one day” portfolio performance					
Mean excess return	0.00745	0.00895	0.00795	0.00715	0.00410
Standard deviation	0.02101	0.01527	0.01438	0.01577	0.04509
Sharpe Ratio	0.35	0.59	0.55	0.45	0.09
Monthly serial correlation	0.14	0.24	0.19	0.12	0.05
Factor model: Fama–French, Momentum, Reversal					
Intercept	0.00545 (3.81)	0.00764 (7.08)	0.00714 (8.66)	0.00512 (5.30)	
Market	−0.06661 (−1.03)	−0.03155 (−0.64)	−0.07697 (−1.77)	−0.14520 (−3.10)	
SMB	−0.04233 (−0.71)	0.00111 (0.02)	−0.02333 (−0.50)	−0.07079 (−1.66)	
HML	0.05740 (1.37)	0.04514 (1.45)	−0.01724 (−0.59)	−0.05403 (−1.82)	
Momentum	−0.02804 (−0.94)	−0.04817 (−2.45)	−0.10312 (−5.83)	−0.18077 (−8.50)	
Reversal	0.10192 (1.50)	0.07237 (1.27)	0.09459 (2.24)	0.20077 (4.34)	
R ²	0.05	0.09	0.18	0.54	
Factor model: Ibbotson factors					
Intercept	0.00716 (6.32)	0.00857 (9.25)	0.00766 (9.39)	0.00651 (7.77)	
Market	−0.00182 (−0.07)	0.01377 (0.74)	0.01642 (0.90)	0.06466 (1.98)	
Small stock premium	0.04120 (1.32)	0.05227 (2.22)	0.03646 (1.66)	0.07608 (1.93)	
Bond default premium	0.14593 (1.11)	0.15989 (1.38)	0.16811 (1.81)	0.30571 (2.82)	
Bond horizon premium	0.07997 (1.55)	0.06818 (1.64)	0.04034 (1.04)	0.03422 (0.77)	
R ²	0.02	0.05	0.04	0.15	

Monthly risk exposures for portfolios of pairs formed and traded according to the “wait one day” rule discussed in the text, over the period between June 1963 and December 2002. The five actors are the three Fama–French factors, Carhart’s Momentum factor, and the Reversal factor discussed in the text. Returns for the portfolios are in excess of the riskless rate. S&P 500 returns are calculated in excess of Treasury bill returns. The Ibbotson factors are from the Ibbotson EnCorr analyzer: The U.S. Small stock premium is the monthly geometric difference between small-company stock total returns and large-company stock total returns. U.S. bond default premium is the monthly geometric difference between total return to long-term corporate bonds and long-term government bonds. The U.S. bond horizon premium is the monthly geometric difference between investing in long-term government bonds and U.S. Treasury bills. The *t*-statistics are in parentheses next to the coefficients and are computed using Newey–West standard errors with six lags.

Table 5
Value at risk (VAR) of pairs trading

	Top 5	Top 20	20 after 100	All
A. Monthly var				
Mean excess return	0.00745	0.00895	0.00795	0.00715
Standard deviation	0.02101	0.01527	0.01438	0.01577
Serial correlation	0.14	0.24	0.19	0.12
VAR				
1%	−0.04320	−0.01943	−0.02236	−0.01994
5%	−0.02142	−0.01002	−0.01293	−0.00877
10%	−0.01516	−0.00577	−0.00756	−0.00614
25%	−0.00460	0.00054	−0.00145	−0.00146
Probability of return below 0	0.35	0.23	0.28	0.32
Minimum historical observation	−0.12628	−0.08218	−0.04266	−0.02951
B. Daily var				
Mean excess return	0.00033	0.00040	0.00035	0.00027
Standard deviation	0.00492	0.00296	0.00277	0.00169
Serial correlation	−0.12	−0.06	0.00	0.35
VAR				
1%	−0.01236	−0.00647	−0.00653	−0.00327
5%	−0.00710	−0.00398	−0.00400	−0.00202
10%	−0.00504	−0.00293	−0.00288	−0.00149
25%	−0.00239	−0.00133	−0.00130	−0.00071
Probability of return below 0	0.47	0.44	0.45	0.46
Minimum historical observation	−0.10079	−0.06723	−0.01987	−0.01069

Monthly and daily VAR percentiles of pairs trading strategies between July 1963 and December 2002 (474 months). Pairs are formed over a 12-month period according to a minimum-distance criterion, and then traded over the subsequent 6-month period. We trade according to the rule that opens a position in a pair on the day following the day on which the prices of the stocks in the pair diverge by two historical standard deviations. The “top n ” portfolios include the n pairs with least distance measures, and the portfolio “101–120” includes the 20 pairs after the top 100. The average number of pairs in the all-pair portfolio is 2057.

momentum factor in our risk model that is constructed along the lines of Carhart (1997).

Table 4 demonstrates that only a small portion of the excess returns of pairs trading can be attributed to their exposures to the five risk factors. The intercepts of the regressions show that risk-adjusted returns are significantly positive and lower than the raw excess returns by about 10–20 bp per month. Because pairs strategies are market-neutral, the exposures to the market are small and with one exception insignificant. Exposures to the other two Fama–French factors, the difference between small and big stocks (SMB) and the difference between value and growth stocks (HML), are not significant and the point estimates alternate in sign. The exposures to momentum and reversals have the predicted signs, and more than half are statistically significant. As can be expected, some of the winner stocks that a pairs strategy shorts are short-term winners, whereas others are medium-term winners. Similarly, the losers are evenly divided between short-term and medium-term losers. Overall the exposures are not large enough to explain the average returns to pairs trading.

The significance of the risk-adjusted returns indicates that pairs trading is fundamentally different from simple contrarian strategies based on reversion [Jegadeesh (1990) and Lehmann (1990)]. The next sections provide further evidence to support this view.

The bottom of Table 4 reports the results of regressing the pairs returns on an alternative set of risk factors suggested by Ibbotson: the excess return on the S&P 500, the U.S. small stock premium, the U.S. bond default premium, and the U.S. bond horizon premium. Although the risk exposures are generally not significant, the signs of coefficients are positive across all portfolios. Thus, when corporate bonds increase in price relative to government bonds, the pairs portfolios make money. There are a range of possible explanations for this pattern. Relatively cheaper borrowing rates by arbitrageurs may force stock prices closer to equilibrium values, or common factors affecting convergence in both stock and bond markets may be responsible. The portfolios also appear to be sensitive to shifts in the yield curve, that is, when long-term spreads decrease, pairs trading is more profitable. At first glance, the sensitivity to term-structure measures may be explained by the presence of interest rate sensitive Utility stocks in many of the top pairs. However, interest rate movements also seem to matter for the more broadly diversified pairs portfolios. In sum, the pairs portfolios seem to have low exposure to various sources of systematic risk. The *R*-squared of most regressions is low, indicating that the portfolios are nearly factor-neutral. This may be expected because they are constructed in a way that should essentially match up economic substitutes.

Table 5 reports both monthly and daily VAR measures to summarize the quantiles of the empirical distributions of the pairs' excess returns. Panel A shows that the worst monthly loss over the almost 40-year sample period was 12.6% for the top five pairs portfolio, and 8.2% for the top 20 portfolio. On average, only once in every hundred months did these portfolios lose more than 4.32 and 1.94%, respectively. Panel B shows that on its worst day, the top five portfolio recorded a 10.08% loss, compared to a 6.72% loss of the top 20 portfolio. On average, only once every hundred days did these portfolios lose more than 1.24% and 0.65%, respectively.

The VAR is useful because it provides a gauge to the potential leverage that could be applied to these strategies. A five-to-one leverage ratio applied to the top five pairs would appear to have been adequate to cover the worst monthly loss in the 40-year period. Although the lessons of recent history have taught us not to rely too heavily on historical VAR measures for gauging capital needs for exploiting convergence strategies, the pairs portfolios seem to be exposed to relatively little risk.

Figure 2 shows the monthly performance of the top 20 pairs, based on the next-day trading rule. Pairs trading was very profitable in the 1970s

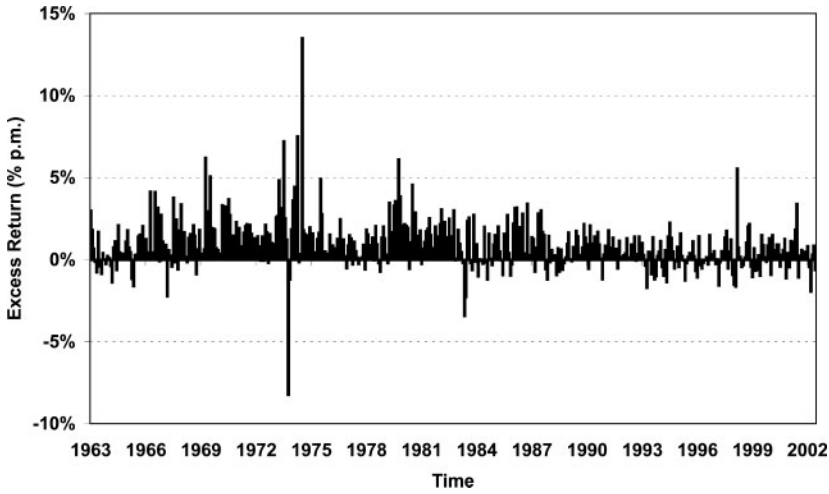


Figure 2
Monthly excess returns of top 20 pairs portfolio
May 1963–December 2002.

and 1980s, and then had a span of more modest performance, when the returns were sometimes negative. Figure 3 compares the cumulative excess returns of the top 20 one-day-waiting strategy with the cumulative excess returns of investment in the S&P 500 index. The smooth index of the pairs trading portfolio contrasts dramatically with the volatility of the

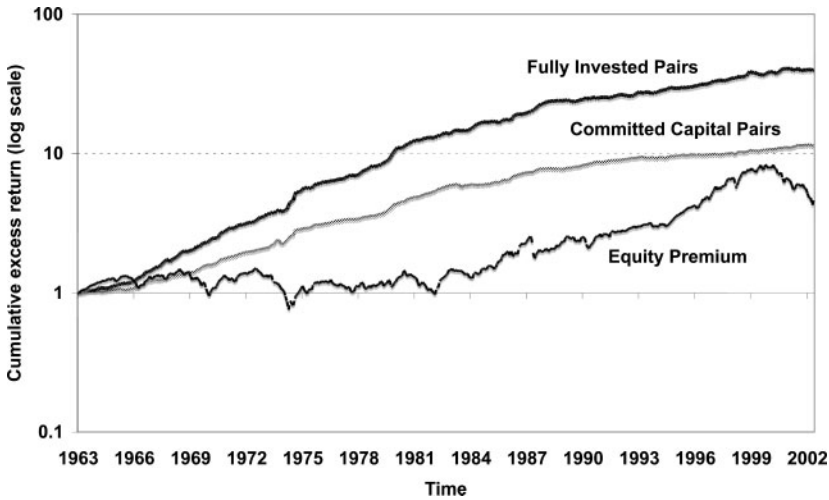


Figure 3
Cumulative excess return of top 20 pairs and S&P 500
May 1963–December 2002.

stock market. Pairs trading performed well over difficult times for U.S. stocks. When the U.S. stock market suffered a dramatic real decline from 1969 through 1980, the pairs strategy experienced some of its best performance. By contrast, in the mid-1990s the market performed exceptionally well, but pairs trading profits were relatively flat. Perhaps after its discovery in the early 1980s by Tartaglia and others, competition decreased opportunity. On the other hand, pairs trading might simply be more profitable in times when the stock market performs poorly.

Two additional explanations may account for the temporal variation in profitability. The first is the long-term trend in transaction costs over the history of our sample. The early part of the analysis represents a period with high fixed commissions. These frictions might have prevented the rapid convergence of relative prices. The secular decrease in transaction costs may have attracted more relative-value equity arbitrage. This is becoming increasingly so with the introduction of more sophisticated trading technologies and networks. The second explanation may be the rise in hedge funds in the period since 1989. The TASS hedge fund database reports that hedge fund assets rose from \$4 billion in 1977 to \$137 billion in 2000, with the most dramatic growth after 1992. Although this database does not capture proprietary trading operations of investment banks, it does suggest that risk arbitrage activities have grown significantly in the last decade of our sample period. Of the \$137 billion in hedge fund assets in 2000, \$119 billion employed “Market Neutral” “Relative Value” or “Arbitrage” strategies, all of which may potentially impact pairs trading operations. Thus, it is possible that lower transaction costs and large inflows of investment capital to relative-value arbitrage have together decreased the rate of return on pairs trading, at least in the simple form we test in this article. In Section 3.8, we will revisit this issue and show that in contrast to the raw returns the risk-adjusted return to pairs trading has been relatively stable over time.

3.6 Pairs trading and contrarian investment

Because pairs trading bets on price reversals, it is an example of a contrarian investment strategy. The results of Table 4 showed that the returns to pairs are positively correlated with—but not explained by—short-term reversals documented by Lehmann (1990) and Jegadeesh (1990). In this section we further explore whether our pairs trading strategies are merely a disguised way of exploiting these previously documented negative autocorrelations. In particular, we conduct a bootstrap where we compare the performance of our pairs to random pairs. The starting point of the bootstrap is the set of historical dates on which the various pairs open. In each bootstrap we replace the actual stocks with two random securities with similar prior one-month returns as the stocks in the actual pair. Similarity is defined as coming from the same decile of previous month’s

Table 6
Returns to random pairs sorted on prior one-month return

Portfolio	Top 5	Top 20	Top 100–120	All 120
A. No waiting				
Fully invested				
Mean excess return	−0.00137	−0.00111	−0.00105	−0.00113
Standard deviation	0.05521	0.02295	0.02264	0.01200
Median	−0.00192	−0.00153	−0.00156	−0.00162
Committed capital				
Mean excess return	−0.00083	−0.00077	−0.00089	−0.00091
Standard deviation	0.02635	0.01358	0.01443	0.00760
Median	−0.00123	−0.00100	−0.00123	−0.00122
B. Wait one day				
Fully invested				
Mean excess return	−0.00177	−0.00004	−0.00154	−0.00156
Standard deviation	0.05966	0.05310	0.02404	0.01213
Median	−0.00241	−0.00172	−0.00206	−0.00201
Committed capital				
Mean excess return	−0.00103	−0.00094	−0.00112	−0.00112
Standard deviation	0.02811	0.01363	0.01449	0.00761
Median	−0.00152	−0.00118	−0.00148	−0.00139

Bootstrap of random pairs traded according to the rule that opens a position in a random pair when the stocks in the true pair diverge by two historical standard deviations and closes the position after the next crossing of prices. The random stocks are selected within the same last-month performance decile on the day the position is opened. The top panel gives summary statistics of the monthly excess returns on value-weighted portfolios of n pairs of stocks where the position is opened immediately. The last column is a portfolio of all 120 top pairs. The bottom panel summarizes the performance with one day waiting before the position is opened. The statistics are computed over 200 replications of the bootstrapped sample.

performance. The difference between the actual and the simulated pairs returns provides an indication of the portion of our pairs return that is not due to reversion. We bootstrapped the entire set of trading dates 200 times. The results are summarized in Table 6. On average we find that the returns on the bootstrapped pairs are well below the true pairs returns. In fact, the excess returns to the simulated pairs are slightly negative, and the standard deviations of the returns are large relative to the true pairs, which is a reflection of the fact that the simulated pairs are poorly matched. The conclusion from the simulations confirms the conclusion from the factor regressions that the pairs strategy does not merely reflect one-month mean reversion. In addition, in the next section, we will show that the long and short portfolios that make up a pair do not provide equal contributions to the profitability of the strategy. These three findings combined strongly suggest that our pairs trading strategy seems to capture temporal variation in returns that is different from simple mean reversion.

3.7 Risk and return of the long and short positions

There are at least three reasons to separately examine the returns to the long and short portfolios that make up a pairs position. First, the

separate returns provide further insight into the question of mean reversion. If pairs trading simply exploits mean reversion, one would expect the abnormal returns to the long and short positions to be equal because the opening of a pair is equally likely to be triggered by either stock. Second, if the excess returns are predominantly driven by the short position, it becomes important to examine whether short-sale considerations might prevent arbitrageurs from competing away the profits. Finally, the risk exposures of the two portfolios can provide further clues to the reason for the profitability of pairs—for example, the possibility that the long and short portfolios have different exposures to common nonstationary risk factors such as bankruptcy risk.⁸ The returns and the risk exposures of the component portfolios are summarized in Table 7.

The table demonstrates that much of the pairs risk-adjusted excess return comes from the short portfolio, which contains the stocks that have increased in value relative to their counterparts prior to opening of the pair. By contrast, the alphas of the long portfolio containing the stocks that decreased in value relative to their counterparts are smaller, and insignificantly different from zero for the top 5 and top 20 portfolios. The asymmetry of the results provides further evidence that the returns to pairs are not due to simple one-month mean reversion. And because much of the abnormal return comes from the short position, which experienced an increase in relative value prior to opening, it is unlikely that the returns are driven by a reward for unrealized bankruptcy risk. The robustness of our results to the cost of short sales will be discussed in Section 3.9.

3.8 Subperiod analysis and the presence of a dormant risk factor

Table 8 summarizes the profitability of pairs trading when we split the sample period at the end of 1988. A comparison between the two top halves of the two panels shows a drop in the raw excess returns to pairs trading. For example, the excess return of the top 20 strategy drops from 118 bp per month to about 38 bp per month. Has increased hedge fund activity arbitrated away the anomalous behavior of pairs since the pre-1989 period? Inspection of the risk-adjusted returns shows that this is not the case: the average risk-adjusted return of the top 20 portfolio drops by about one-third from 67 to 42 bp per month, but remains significantly positive in both subperiods ($t = 4.41$ and 3.77 , respectively). Changes in the factor exposures and factor volatilities can explain only part of the lower returns in the early part of the sample, but not the risk-adjusted returns to pairs trading.

Are the positive risk-adjusted returns to pairs trading a general failure of our risk model? In other words, are there reasons to believe that the

⁸ We thank a referee for suggesting this explanation.

Table 7
Returns to long and short components of pairs

Portfolio	Top 5		Top 20		20 after top 100		All	
	Long	Short	Long	Short	Long	Short	Long	Short
Portfolio performance								
Mean monthly return	0.01245	0.00501	0.01330	0.00435	0.01458	0.00663	0.01623	0.00908
Standard error (Newey-West)	0.00183	0.00177	0.00179	0.00174	0.00170	0.00165	0.00254	0.00231
Standard deviation	0.03875	0.03259	0.03653	0.03161	0.03437	0.03146	0.05157	0.04601
Monthly serial correlation	0.06	0.11	0.09	0.12	0.13	0.12	0.17	0.16
Regression on Market, SMB, HML, Momentum, and Reversal factors								
Intercept	0.00103	-0.00442	0.00243	-0.00521	0.00287	-0.00426	-0.00101	-0.00613
<i>t</i> -Statistic	0.47	-2.74	1.27	-3.35	2.85	-4.26	-1.32	-5.22
U.S. equity risk-premium	0.37415	0.44075	0.47617	0.50772	0.48520	0.56217	0.58571	0.73091
<i>t</i> -Statistic	4.60	6.19	6.68	8.44	10.23	9.74	14.99	12.13
SMB	-0.16764	-0.12532	-0.06506	-0.06616	0.01120	0.03453	0.22192	0.29271
<i>t</i> -Statistic	-1.74	-2.00	-0.80	-1.22	0.27	0.95	3.45	3.15
HML	0.48401	0.42661	0.52539	0.48025	0.39451	0.41175	0.25732	0.31134
<i>t</i> -Statistic	7.12	6.92	9.36	9.47	9.76	11.54	7.49	8.11
Momentum	-0.05430	-0.02625	-0.04456	0.00361	-0.10901	-0.00589	-0.08832	0.09245
<i>t</i> -Statistic	-1.48	-0.65	-1.21	0.10	-3.53	-0.21	-3.53	2.81
Reversal	0.15854	0.05662	0.09601	0.02365	0.17288	0.07830	0.39208	0.19131
<i>t</i> -Statistic	1.83	1.13	1.26	0.55	4.33	1.90	10.44	2.98
R^2	0.39	0.41	0.48	0.49	0.78	0.75	0.97	0.93

Monthly risk profile for the long and short positions of the pairs portfolios formed and traded according to the “wait one day” rule discussed in the text. The returns in the bottom half of the table are in excess of the 30-day Treasury bill returns. The risk adjustment includes the Fama–French factors, as well as Momentum and the Reversal factors discussed in the text. The *t*-statistics are computed using Newey–West standard errors with six lags. Absolute kurtosis is reported.

Table 8
Subperiod analysis

Portfolio	Top 5	Top 20	20 after top 100	All	Factor SD
A. Pre-1989					
“Wait one day” portfolio performance					
Mean excess return	0.01034	0.01181	0.01052	0.00992	
Standard deviation	0.02259	0.01689	0.01527	0.01651	
Regression on Fama–French factors					
Intercept	0.00353	0.00670	0.00710	0.00446	
<i>t</i> -Statistic	1.72	4.41	6.54	4.19	
U.S. equity risk-premium	−0.43395	−0.31200	−0.28946	−0.43429	0.04580
<i>t</i> -Statistic	−4.29	−3.57	−3.80	−7.26	
SMB: small minus big	−0.44181	−0.33193	−0.27508	−0.40184	0.02923
<i>t</i> -Statistic	−3.75	−2.86	−2.86	−5.76	
HML: high minus low book to market	0.03568	0.03162	−0.06840	−0.06983	0.02597
<i>t</i> -Statistic	0.64	0.73	−1.84	−1.97	
Momentum	0.01291	−0.01630	−0.07689	−0.15848	0.03506
<i>t</i> -Statistic	0.29	−0.50	−2.92	−5.01	
Reversal	0.43575	0.33274	0.28765	0.45222	0.07228
<i>t</i> -Statistic	4.44	3.62	4.05	7.98	
<i>R</i> ²	0.19	0.23	0.26	0.70	
B. Post-1988					
“Wait one day” portfolio performance					
Mean excess return	0.00217	0.00375	0.00327	0.00212	
Standard deviation	0.01660	0.00987	0.01121	0.01295	
Regression on Fama–French factors					
Intercept	0.00337	0.00417	0.00363	−0.00065	
<i>t</i> -Statistic	2.12	3.77	3.01	−0.56	
U.S. equity risk-premium	0.07339	0.04804	0.00241	−0.06958	0.04390
<i>t</i> -Statistic	1.68	1.81	0.06	−2.43	
SMB: small minus big	−0.00400	0.02888	0.02332	−0.06063	0.03856
<i>t</i> -Statistic	−0.10	1.32	0.69	−2.83	
HML: high minus low book to market	0.03441	0.01412	0.01830	−0.08202	0.03641
<i>t</i> -Statistic	0.61	0.45	0.50	−2.65	
Momentum	−0.00424	−0.02266	−0.08840	−0.12670	0.04926
<i>t</i> -Statistic	−0.13	−1.29	−4.17	−7.59	
Reversal	−0.06259	−0.01727	0.02404	0.18808	0.04448
<i>t</i> -Statistic	−1.76	−0.67	0.54	5.56	
<i>R</i> ²	0.02	0.04	0.19	0.64	

Monthly risk profile for portfolios of pairs formed and traded according to the “wait one day” rule discussed in the text, over the two subperiods between July 1963 and December 1988 (Panel A) and between January 1989 and December 2002 (Panel B). The “top *n*” portfolios include the *n* pairs with least distance measures, and the portfolio “20 after top 100” includes the 20 pairs after the top 100 pairs. The average number of pairs in the all-pair portfolio is 2057. The *t*-statistics are computed using Newey–West correction with six lags for the standard errors.

risk-adjusted pairs returns are a compensation for an omitted (latent) risk factor? Inspection of the correlation between disjoint pairs portfolios provides some support for this view. Moreover, this latent risk factor seems to have been relatively dormant over the second half of our sample, which can account for the lower recent profitability of pairs trading. The full-sample correlation between the excess returns of the top 20 and the 100–120 pairs portfolios is 0.48. Because there is no overlap between the positions of these portfolios, the correlation indicates the presence of a

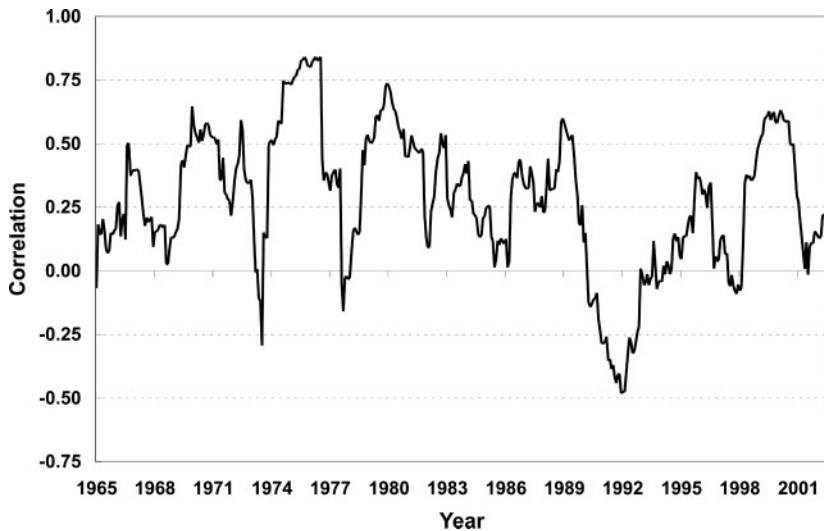


Figure 4
Top 20 pairs and 101–120 pairs portfolios
 Rolling 24-month correlation of five-factor risk-adjusted excess returns.

common factor to the returns. Moreover, the correlation is 0.51 over the profitable pre-1988 period but much lower (0.18) over the lower excess return post-1988 period. These correlations are not driven by the 5 “systematic” risk factors we considered, because the correlations of the residuals from the factor regressions are very similar to the raw correlations. In particular, the correlation between the top 20 Fama–French–Momentum–Reversal (FFMR) residuals and the top 100–120 FFMR residuals is 0.41. The respective subperiod FFMR correlations are 0.42 and 0.20. This is further illustrated in Figure 4, which shows that the rolling 24-month correlation between the two portfolios was especially high during the pre-1989 period.

These results suggest that there is common component to the profits of pairs portfolios that is not captured by our conventional measures of systematic risk. The common component was stronger during the first half of our sample, which is consistent with higher profits (abnormal returns) than in the second half of our sample when the factor was more dormant. These results are consistent with the view that the abnormal returns documented in this article are indeed a compensation for risk, in particular the reward to arbitrageurs for enforcing the “Law of One Price.”

3.9 Robustness to short-selling costs

Having identified and back-tested a filter rule, and subjected it to a range of controls for risk, the question of the nature of pairs profits remains. Why do prices of close economic substitutes diverge and converge? The

convergence is easier to understand than the divergence, given the natural arbitrage motivation and the documented existence of relative-value arbitrageurs in the U.S. equity market. However, this does not explain why prices drift away from parity in the first place. One possible explanation is that prices diverge on random liquidity shocks that cannot be exploited by professional arbitrageurs because of short-selling costs.⁹

First, there are explicit short-selling costs in the form of specials. Second, D'Avolio (2002) argues that short recalls are potentially costly because they may deprive arbitrageurs of their profits. This opportunity cost is reinforced by D'Avolio's findings that short recalls are more common as the price declines. For example, if the short stock is recalled when it starts to converge downward, then the pairs position is forced to close prematurely and the arbitrageur does not capture the profit from the pair convergence. We perform two tests for robustness of the profits that corresponds to these two types of short-selling costs.

The first test is motivated by the findings of D'Avolio (2002) and Geczy et al. (2002) that specials have minimal effect of large stocks. Correspondingly, we test for robustness of profits by trading pairs that are formed using only stocks in the top three size deciles.¹⁰ The results, given in Panel A of Table 9, can be directly compared to those in Panel B of Table 1. The comparison shows that the profits of the top 20 strategy drop by about 2 bp per month, but increase for the top five portfolios. Overall, the profits change little and remain highly significant. This shows that the pairs trading profits are not driven by illiquid stocks that are likely to be on special.

The second test is motivated by the evidence in Chen et al. (2002) and D'Avolio (2002) that short recalls are driven by dispersion of opinion. We use high volume as proxy for divergence of opinion and perform pairs trading under recalls, where we simulate recalls on the short positions, and subsequent closing of the pairs position, on days with high volume. High-volume days are defined as days on which daily volume exceeds average daily volume over the 18 months (both split-adjusted) by more than one standard deviation. Panel B of Table 9 summarizes the profits of pairs trading with the high-volume recalls. The profits decline slightly by 4–13 bp per month, yet they remain large and positive. For example, the top 20 pairs portfolio earns an average of 85 bp per month with short recalls, with a Newey-West *t*-statistic of 9.07. The results in Panel B can be interpreted as an estimate of the opportunity cost of short recalls.

⁹ We are grateful to the editor Maureen O'Hara for pointing out this plausible explanation and for suggesting the way to test it.

¹⁰ Using large liquid stocks also mitigates the problem of stocks that are hard to short and can be overvalued, see, for example, Jones and Lamont (2002).

Table 9
Robustness to short-selling costs

Portfolio	Top 5	Top 20	20 after top 100	All
A. Top three deciles				
Mean monthly return (fully invested)	0.00835	0.00914	0.00728	0.00690
Standard error (Newey-West)	0.00112	0.00096	0.00075	0.00064
<i>t</i> -Statistic	7.47	9.48	9.66	10.85
Excess return distribution				
Median	0.00800	0.00793	0.00614	0.00508
Standard deviation	0.01904	0.01464	0.01579	0.01294
Kurtosis	0.02	0.98	0.80	1.77
Skewness	7.01	8.20	6.46	11.67
Minimum	-0.10894	-0.05646	-0.03803	-0.02680
Maximum	0.10144	0.10003	0.10642	0.10478
Observations with excess return < 0	32%	26%	33%	27%
Mean excess return on committed capital	0.00493	0.00514	0.00462	0.00401
B. Short recalls on high volume				
Mean monthly return (fully invested)	0.00619	0.00854	0.00665	0.00585
Standard error (Newey-West)	0.00115	0.00094	0.00085	0.00098
<i>t</i> -Statistic	5.39	9.07	7.81	5.95
Excess return distribution				
Median	0.00592	0.00703	0.00531	0.00312
Standard deviation	0.02210	0.01488	0.01505	0.01555
Skewness	0.15	0.48	1.03	2.82
Kurtosis	5.39	6.00	7.39	18.98
Minimum	-0.09611	-0.06594	-0.04408	-0.03465
Maximum	0.10504	0.07255	0.10084	0.14173
Observations with excess return < 0	39%	26%	34%	37%
Mean excess return on committed capital	0.00304	0.00347	0.00296	0.00202

Summary statistics of the monthly excess returns on portfolios of pairs. We trade according to the rule that opens a position in a pair when the prices of the stocks in the pair diverge by two historical standard deviations. Panel B reports the summary statistics for the rule that waits one day before opening and closing the position. The “top *n*” portfolios include the *n* pairs with least distance measures, and the portfolio “20 after top 100” includes the 20 pairs after the top 100 pairs. The average number of pairs in the all-pair portfolio is 2057. There are 474 monthly observations, from July 1963 through December 2002. The *t*-statistics are computed using the Newey-West standard errors with six-lag correction. Absolute kurtosis is reported.

Overall, the small effects confirm that the profits persist when trading pairs of large stocks as well as when shorts are recalled. These results show that pairs trading profits are robust to short-selling costs. For better-positioned investors, for example, large institutions and hedge funds, the pairs trading profits are likely to remain essentially unaffected by potential shorting costs. Getczy et al. (2002) argue that for large traders, who have better access to most stocks at “wholesale prices,” direct shorting costs of the rebate rate on short sales are low (4–15 bp/year). The main implicit shorting cost stems from limited availability and is relevant mostly for general retail investors. The impact of such potential short sales constraints on the profitability of pairs trading by large investors is mitigated by our use of liquid stocks that trade every day over a period of one year.

4. Conclusion

We examine a hedge fund equity trading strategy based on the notion of co-integrated prices in a reasonably efficient market, known on Wall Street as pairs trading. Pairs are stocks that are close substitutes according to a minimum-distance criterion using a metric in price space. We find that trading suitably formed pairs of stocks exhibits profits, which are robust to conservative estimates of transaction costs. These profits are uncorrelated to the S&P 500; however, they do exhibit low sensitivity to the spreads between small and large stocks and between value and growth stocks in addition to the spread between high- and intermediate-grade corporate bonds and shifts in the yield curve. In addition to risk and transaction cost, we rule out several explanations for the pairs trading profits, including mean reversion as previously documented in the literature, unrealized bankruptcy risk, and the inability of arbitrageurs to take advantage of the profits because of short-sale constraints.

One view of the lower profitability of pairs trading in recent year is that returns are competed away by increased hedge fund activity. The alternative view, taken in this article, is that abnormal returns to pairs strategies are a compensation to arbitrageurs for enforcing the “Law of One Price.” We present two pieces of empirical evidence that supports this view. First, although raw returns have fallen, the risk-adjusted returns have continued to persist despite increased hedge fund activity. Second, our results suggest that the change in risk-adjusted returns of pairs trading is accompanied by the diminished importance of a common factor that drives the returns to pairs strategies. A further examination of the nature of this common factor and its link to the profitability of pairs trading is an important question for future research.

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